



The Upside Down Labs team

sensors and multiple accessories, all catering to bio-physiological and biopotential signals.

The company claims to be India's first and only one to design and manufacture DIY neuroscience kits. "The uniqueness of these kits lies in their fully open source nature—every design file is available for anyone to study, modify, or build upon. This makes them particularly valuable for students developing their own biosensors," Deepak explains.

Discussing the electronics involved, Deepak says, "The electronics in each sensor are designed from scratch. Most use a quad-operational amplifier. The first stage is a custom instrumentation amplifier that processes the signal, followed by another amplifier for band-pass filtering, and a fourth amplifier that creates the reference voltage power supply. This setup forms the core of most BioAmp designs. An earlier version, BioAmp v1.5, had a different design, but all current models follow

this structure: two amplifiers for signal processing, one for filtering, and one for power supply."

On revenue, Deepak states, "We have sold our products in over 60 countries and have distributors in several regions. While we cannot disclose exact revenue figures, we have sold over 5000 DIY kits alone."

Everything is handled in-house except PCB manufacturing, which is outsourced. The process begins with designing the PCB layout and placing components. Once finalised, the design is sent for manufacturing. After production, testing is conducted, and any issues are resolved. The design is then refined and re-tested before being approved for larger-scale assembly using an in-house pick-and-place machine.

Discussing the challenges of designing a bio amplifier sensor, Deepak says, "Capturing weak biopotential signals like EEG, EMG, and ECG—typically in the microvolt or millivolt range—requires precise amplification without distortion. Electrical noise and interference from power sources, surrounding electronics, and body movements further complicate signal accuracy, making effective filtering and shielding essential."

Regarding the target users of the bio amplifier sensor, Deepak reveals, "Researchers have used them for

material science, designing new electrodes, and testing conductive materials. Students use these sensors for science fairs and personal research. A company called Incognito Blueprints also incorporates these kits to help students build science projects and portfolios, enhancing their chances of getting into top universities like Ivy League colleges."

The company is entirely self-funded, reinvesting revenue from product sales into research and development. Currently, it faces no major challenges and continues to expand, developing more advanced products. With ongoing recruitment, the team has grown from just five members to nearly 15.

The company is now primarily focused on enhancing software support for its biopotential amplifiers. Additionally, they are working on new products and will be integrating wireless technology integration with Neuro Play Ground Lite, a multi-channel wireless biopotential signal amplification device designed for recording EMG, ECG, EOG, and EEG. This device will be soon launched on Crowd Supply.

Looking ahead, the startup aims to design new products, advance technology, and promote open source neuroscience. It seeks support in building distribution networks and partnering with academic institutions to organise neuroscience workshops. Moving forward, their goal remains clear: continuous innovation and sustained growth.

—Nidhi Agarwal

3 ACCELERATING INDIA'S AI ASPIRATIONS WITH EXSLERATE

Nvidia may have done it first, but that does not mean India will stay behind. That is why Bengaluru-based startup SandLogic has immersed itself in developing an AI-based co-processor IP, ExSLerate.

The name SandLogic comes from the two basic building blocks of a semiconductor processor chip—silicon and intellect. The popularisation of artificial